

Week-01: GitHub Workflow Tutorial

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Working with course-hub and your personal tutorial repository tutorials-exercises.

Estimated time: 60-75 minutes

Background

In this course, you will use two different GitHub repositories for in-class work (you will use others for homework assignments):

1. **course-hub** contains course materials provided by the instructor.
2. **Your personal tutorial repository** contains the tutorials and exercises you complete during class.

You will **pull** new materials from **course-hub**, but you will complete, save, commit, and push your work in your personal tutorial repository.

Learning outcomes

By the end of this tutorial, you will be able to:

- distinguish between the central course repository and your personal tutorial repository;
- explain the difference between a remote GitHub repository and a local repository on your computer;
- clone a repository from GitHub to your computer;
- pull updates from GitHub;

- copy course materials into the correct repository;
- review, stage, commit, and push changes;
- confirm that your work appears on GitHub; and
- reopen an existing RStudio Project without cloning it again.

1. Create a course folder on your computer

Call it `skill-building-in-R/`. This folder will house many GitHub repositories - the course-hub, the tutorials-exercises (on your personal account), and your homework repositories. For now, we will focus on the course-hub and the tutorials-exercises repositories.

`skill-building-in-R/` is your parent folder.

2. Understand the two repositories

Central course repository: `course-hub`

The instructor controls this repository. It contains template files for the tutorials and exercises.

```
skill-building-in-R/  
  course-hub/  
    course-hub.Rproj  
    README.md  
    week-01/  
      tutorials/  
      exercises/  
    week-02/
```

A trailing `/` indicates that the item is a **folder**. In the above folder structure, `skill-building-in-R/`, `course-hub/`, `week-01/`, and `week-02/` are all folders, not files. Files will have an extension, like `README.md`.

`course-hub.Rproj` is an Rproject file, something we will learn about next week.

`README.md` is a file - every repository (`course-hub`) should have a `README` file, which states what the repository holds. The extension `.md` just means it is a markdown file. A markdown file is a plain text file that uses a human-readable syntax to format text. We will learn more about markdown (and its newer cousin, quarto) next week.

In the `week-01/` and `week-02/` folders (and for every week), there are subfolders for the tutorial templates and the exercises templates.

In `course-hub`, you will:

- clone the repository once;
- pull updates before class;
- view course materials; and
- copy the tutorial and exercise files you need.

You will **not** complete or save your work in `course-hub`, and you do not have the ability to push changes to it.

Personal tutorial repository

You will create one personal repository for all in-class tutorials and exercises during the semester. This will become a foundation of code you can refer back to later.

You can name your tutorials-exercise repo:

```
tutorials-exercises
```

Your repository might eventually look like this:

```
skill-building-in-R/  
  tutorials-exercises/  
    README.md  
    week-01/  
      tutorials/  
      exercises/  
    week-02/  
      tutorials/  
      exercises/  
    tutorials-exercises.Rproj
```

In your personal tutorial repository, you will:

- copy tutorial and exercise templates from `course-hub`;
- complete tutorials and exercises;
- save your work;
- commit your changes; and
- push your work to GitHub.

I will not be looking at or grading tutorials or exercises. These are for your own edification. However, your participation points will reflect your engagement during both.

3. Remote and local repositories

A repository on GitHub is the **remote** copy. The copy on your computer is the **local** copy.

```
GitHub
remote repository stored online
    ↓ clone or pull
Your computer
local repository opened in RStudio
    ↑ push
```

Important terms:

Term	Meaning
Clone	Download a repository to your computer for the first time
Pull	Download new changes into an existing local repository
Commit	Save a checkpoint in the repository history on your computer
Push	Upload your local commits to GitHub

You clone a repository only once per computer. After that, you reopen the existing project and pull updates.

4. Organize your course folders

Your repositories should be in the parent folder (next to one another), not inside one another.

Recommended structure:

```
skill-building-in-R/
  course-hub/
  tutorials-exercises/
  homeworks
    homework-01-lastname/
    homework-02-lastname/
```

Do **not** place one Git repository inside another.

Incorrect:

```
course-hub/  
  tutorials-exercises/
```

Correct:

```
skill-building-in-R/  
  course-hub/  
    tutorials-exercises/
```

5. Clone course-hub

You only need to do this once on each computer you use.

On GitHub

1. Open the [course-hub repository link](#)
2. Click the green **Code** button.
3. Make sure **HTTPS** is selected.
4. Copy the repository URL.

In RStudio

1. Open RStudio.
2. Select **File** → **New Project**.
3. Select **Version Control**.
4. Select **Git**.
5. Paste the repository URL into **Repository URL**.
6. Choose your `skill-building-in-R/` parent folder as the project directory.
7. Click **Create Project**.

RStudio will clone the repository and open the project. A course-hub folder will automatically be created through this process.

Check your work

Confirm that:

- `course-hub` appears as the project name in the upper left-hand corner of RStudio;
 - the Files pane shows the contents of `course-hub`;
 - the Git pane is visible;
 - `course-hub` folder was created in `skill-building-in-R/` folder; and
 - you can locate the `week-01` folder.
-

6. Practice pulling an instructor update

Your instructor will make a small change to `course-hub` during class.

When instructed:

1. Go to [course-hub repository link](#) and note there is no file named `test.R`.
2. Make sure the `course-hub` project is open in RStudio.
3. Open the **Git** pane.
4. Click **Pull**.
5. Wait for the pull to finish.
6. Locate `test.R`.
7. Go back to [course-hub repository link](#) and note there is now a `test.R` file.

Key idea

Clone once. Pull whenever the instructor updates the repository (pull at the start of class).

Pulling does not create a second copy of the repository. It updates the local copy you already have.

7. Create your personal tutorial repository

Create this repository under your **personal GitHub account**, not inside `course-hub`. This is where you will copy over the templates for the tutorials and exercises, and complete, save, commit, and push them.

On GitHub

1. Go to GitHub.
 2. Click the + menu in the upper-right corner.
 3. Select **New repository**.
 4. Under **Owner**, select your personal GitHub account (this is likely already selected and you will only have options here if you are part of organizations).
 5. Name the repository:
`tutorials-exercises`
 6. Add a short description, such as:
`In-class tutorials and exercises for Skill Building in R`
 7. Choose a ‘repository visibility’ option. The repository visibility is asking whether you want this to be a private repo (only you and collaborators you add can see it) or public (anyone can see it). I usually keep in-progress analyses private and make them public when I submit the paper to the journal (so reviewers can see code) or a preprint server (so readers can see code). For this class, it does not matter as much - choose what you feel like.
 8. Check **Add a README file** (this is a description of the project, let’s look at some other README files).
 9. Click **Create repository**.
-

8. Clone your tutorial repository

On GitHub

1. Open your new tutorial repository.
2. Click **Code**.
3. Select **HTTPS**.
4. Copy the repository URL.

In RStudio

1. Select **File** → **New Project**.
2. Select **Version Control**.
3. Select **Git**.
4. Paste the repository URL.
5. Choose the same `skill-building-in-R/` parent folder used for `course-hub`.
6. Click **Create Project**.

Your repositories should now be side by side:

```
skill-building-in-R/  
  course-hub/  
  tutorials-exercises/
```

9. Copy the Week 1 materials

You will complete the tutorial in your personal tutorial repository, not in `course-hub` so you will need to copy the tutorial(s) and exercises for the week into your `tutorials-exercises` repository.

Close or save any open files

Before copying files, save your work and close any files you are not using.

Using Finder or File Explorer

1. Open the `course-hub` folder on your computer.
2. Open `week-01`.
3. Copy the `tutorials` and `exercises` folders.
4. Open your personal `tutorials-exercises` folder.
5. Create a folder named:

```
week-01
```

6. Paste the copied materials into `week-01`.

Your personal repository should now look similar to course-hub, which looks like this for week-01:

```
skill-building-in-R/  
  tutorials-exercises/  
    README.md  
    week-01/  
      tutorials/  
        tutorial-01-github.md  
      exercises/  
        exercises-01-github.md
```

10. Open Rstudio as if you are just starting class

When you open RStudio for a working session, you want to open it using the Rproject file.

To practice this:

1. Quit RStudio so it is not open.
2. Navigate to and open your tutorials-exercises/ folder.
3. To open this project (and thus RStudio), click on the tutorials-exercises.Rproj file.
4. Confirm that the project name in the upper left-hand corner of RStudio is tutorials-exercises and your files are there in the file pane.

This is how you open RStudio. I always organize my work into RProjects and so when you click on the .Rproj file, it opens RStudio and your working directory is set to the folder (tutorials-exercises/). We will learn more about the `{here}` package next week which is used in conjunction with Rprojects to solve all working directory issues.

11. Your weekly in-class workflow

At the beginning of each class:

1. Open `course-hub`.
2. Pull the latest instructor updates.
3. Copy that week's tutorial and exercise materials.
4. Paste them into the matching week folder in your `tutorials-exercises` repository.

Then:

5. Open your `tutorials-exercises` RStudio Project.

6. Complete the tutorial and exercises.
7. Save your files.
8. Review your changes in the Git pane.
9. Stage the files you want to commit.
10. Commit with an informative message.
11. Push to GitHub.
12. Verify that your work appears in the remote GitHub repository.

12. Repository rules for this course

In `course-hub`

You should:

- pull updates;
- view course materials; and
- copy starter files.

You should **not**:

- complete your work there;
- commit changes there; or
- push changes there.

In your `tutorials-exercises` repository

You should:

- complete tutorials and exercises;
- save your work;
- review your changes;
- commit meaningful checkpoints; and
- push regularly.

13. One-sentence workflow

Pull course materials from `course-hub`, copy them into your personal `tutorials-exercises` repository, complete your work there, and then review, stage, commit, and push your changes to GitHub.